<table>
<tr><th colspan="4" rowspan="2">Question</th><th rowspan="2">Marking details</th><th colspan="6">Marks available</th></tr>
<tr><th>AO1</th><th>AO2</th><th>AO3</th><th>Total</th><th>Maths</th><th>Prac</th></tr>
<tr><td>3</td><td>(a)</td><td></td><td></td><td>
<table>
<tr><th>Property of the ball</th><th>Stays constant</th><th>Gets smaller</th><th>Gets larger</th></tr>
<tr><td>acceleration</td><td>✓</td><td></td><td></td></tr>
<tr><td>speed</td><td></td><td>✓</td><td></td></tr>
<tr><td>kinetic energy</td><td></td><td>✓</td><td></td></tr>
<tr><td>potential energy</td><td></td><td></td><td>✓</td></tr>
<tr><td>total energy</td><td>✓</td><td></td><td></td></tr>
</table>
1 mark for each correct row</td><td>4</td><td></td><td></td><td>4</td><td></td><td></td></tr>
<tr><td></td><td>(b)</td><td>(i)</td><td></td><td>Substitution: 0.46 × 10 (1)
= 4.6 [N] (1)</td><td>1</td><td>1</td><td></td><td>2</td><td>2</td><td></td></tr>
<tr><td></td><td></td><td>(ii)</td><td></td><td>Substitution: 4.6 (ecf) × 15 (1)
= 69 (1)
Joules or J or N m (1)</td><td>1

1</td><td>1</td><td></td><td>3</td><td>2</td><td></td></tr>
<tr><td></td><td></td><td>(iii)</td><td></td><td>[Yes because]
acceleration [due to gravity] is the same as gravitational field strength
or no other forces acting
or $a = g$
or weight = 0.46 × 10 also
or weight is 4.6 [N]
or gravitational field strength = 10 or g is 10
Accept they’re both 10 or we only multiplied by 10</td><td></td><td></td><td>1</td><td>1</td><td></td><td></td></tr>
<tr><td></td><td></td><td></td><td></td><td>Question 3 total</td><td>7</td><td>2</td><td>1</td><td>10</td><td>4</td><td>0</td></tr>
</table>

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